SOLUTIONS

Self Assessment Paper-5

SECTION A

1. (a) net charge enclosed and permittivity of the medium

Why? Because by Gauss's Law it depends only on net enclosed charge, bell on permittivity not on surface size.

Concept: Gauss' law

2. (d) conservation of charge and conservation of energy

Why? **Kirchhoff's First Law ($\Sigma I = 0$):** At a junction, the sum of currents entering equals the sum of currents leaving — based on conservation of charge (no charge is lost).

Kirchhoff's Second Law ($\Sigma IR = \Sigma E$): The sum of potential drops equals the sum of emfs in a closed loop — based on conservation of energy (total energy gained equals total energy lost).

Concept: Kirchhoff's law

3. (b) II

Why? Applying the right-hand thumb rule, the strongest magnetic field is in region II.

Concept: Right hand thumb rule of magnetism

4. (d) 75 mH

Why? Coefficient of self-inductance for solenoid,

$$L = \frac{\mu_0 N^2 A}{l}$$

So, $$L \propto N^2$$

$$\therefore \quad \frac{L_1}{L_2} = \frac{N_1^2}{N_2^2}$$

Putting values, $$\frac{108}{L_2} = \frac{600^2}{500^2}$$

$$\therefore \quad L_2 = 108 \times \frac{25}{36} = 75 \text{ mH}$$

Concept: Coefficient of self inductance

5. (a) low resistance in parallel

Why? When a high resistance is connected in series with the galvanometer it works as a voltmeter and when a low resistance is connected in parallel with the galvanometer it works as a ammeter.

Concept: Ammeter

6. (a) $\frac{15}{\sqrt{2}}$ A

Why? Instantaneous current,

$$I = I_0 \sin \omega t$$

or $$I = I_{rms} \times \sqrt{2} \times \sin(2\pi f t)$$

Putting values,

$$I = 15 \times \sqrt{2} \times \sin\left(\frac{2 \times \pi \times 50}{600}\right)$$

or $$I = 15 \times \sqrt{2} \times \sin\left(\frac{\pi}{6}\right)$$

$$\therefore \quad I = \frac{15}{\sqrt{2}}$$

Concept: AC current

7. (c) $\frac{2E}{3}$

Why? Given:

- EMF of cell E
- Internal resistance r
- External resistance $R = 2r$

Current through the circuit

$$I = \frac{E}{R+r} = \frac{E}{2r+r} = \frac{E}{3r}$$

Potential drop across the terminals

$$V = I \cdot R = \frac{E}{3r} \cdot 2r = \frac{2E}{3}$$

Concept: EMF of cell

8. (b) $\frac{P}{2}$

Why? Lens maker's formula

$$P = \frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For a symmetric biconcave lens: $R_1 = -R$, $R_2 = R$:

$$P = (n-1)\left(-\frac{1}{R} - \frac{1}{R}\right)$$

$$= (n-1)\left(-\frac{1}{R} - \frac{1}{R}\right) = -\frac{2(n-1)}{R}$$

When split vertically, each half becomes a plano-concave lens with one curved surface of radius R and one plane surface($R = \infty$):

$$P_{half} = (n-1)\left(0 - \frac{1}{R}\right) = \frac{-(n-1)}{R}$$

Compare powers

Original power:

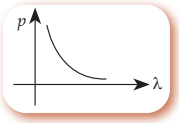
$$P_{\text{original}} = -\frac{2(n-1)}{R}$$

Half-lens power:

$$P_{\text{half}} = -\frac{n-1}{R} = \frac{P_{\text{original}}}{2}$$

Concept: Lens maker's formula

9. (d)



Why? According to de-Broglie,

$$\lambda = \frac{h}{p}$$

So clearly:

$$\lambda \propto \frac{1}{p}$$

Concept: de-Broglie wavelength

10. (d) 1

Why? Nuclear density is independent of mass number.

$$\frac{\rho_{64}}{\rho_{125}} = 1$$

Concept: Nuclear density

11. (a) I

Why? In the Bohr model, the energy of a photon emitted during a transition is given by:

$$E_{\text{photon}} = E_i - E_f$$

where $E_n = -\frac{13.6}{n^2}$ eV for hydrogen .

- I: $n = 4 \rightarrow n = 1$

$$E = -\frac{13.6}{1^2} - \left(-\frac{13.6}{4^2}\right)$$

$$= -13.6 + 0.85 = 12.75 \text{ eV}$$

- II: $n = 4 \rightarrow n = 3$

$$E = -\frac{13.6}{3^2} - \left(-\frac{13.6}{4^2}\right)$$

$$= -1.511 + 0.85 = 0.661 \text{ eV}$$

- III: $n = 3 \rightarrow n = 1$

$$E = -13.6 + 1.511 = 12.089 \text{ eV}$$

- IV: $n = 3 \rightarrow n = 2$

$$E = -3.4 + 1.511 = 1.889 \text{ eV}$$

The largest energy difference is 12.75 eV for transition I.

Concept: Bohr atomic theory

12. (a) Diffusion current is equal to drift current

Why? In an unbiased *p-n* junction, the net current (i.e., the sum of drift current and diffusion current) is zero. So, diffusion current is equal and opposite to the drift current.

Why not others: They incorrectly assume that one or both currents vanish.

Concept: *p-n* junction

13. (a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Why? The assertion is true because in a moving-coil galvanometer, the deflection angle θ is directly proportional to the current I passing through it:

At equilibrium, $k\theta = NBIAk$,

so $\theta \propto I$

The reason is true because the coil of a galvanometer is suspended in a uniform radial magnetic field, which ensures that the plane of the coil remains parallel to the magnetic field lines during rotation.

Reason explains *why* this proportionality holds true — because the coil is placed in a **uniform radial magnetic field**, which ensures the deflecting torque is directly proportional to current for all angular positions of the coil.

Concept: Galvanometer

14. (c) Assertion is true but Reason is false.

Why? Assertion is true because when a bar of copper is placed in an external magnetic field, the field lines are slightly concentrated inside because copper weakly allows magnetic field penetration.

Reason is false because Copper is not paramagnetic; it is diamagnetic, meaning it actually produces a very weak opposing magnetic field. The slight concentration effect described in the assertion is not caused by paramagnetism.

Concept: Magnetism and matter

15. (a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Why? In Young's Double Slit Experiment (YDSE), fringe width is given by:

$$\beta = \frac{\lambda D}{d}$$

Since λ, D and d are constants during the experiment, β remains constant → all fringes have equal width.

Assertion is true because all fringes have the same width because β is constant.

Reason is also true because the expression for β shows dependence on λ, D and d.

Here, reason correctly explains the assertion.

Concept: Young's double slit experiment

16. (c) Assertion is true but Reason is false.

Why? Assertion is true because diffraction is more pronounced when the wavelength of a wave is comparable to or larger than the size of the obstacle or aperture.

Radio waves have much longer wavelengths than infrared waves, so they bend around obstacles more easily, producing a stronger diffraction effect.

The reason is false because radio waves do not have greater frequency than infrared waves; in fact, they have much lower frequency and longer wavelength.

Diffraction depends on wavelength, not on frequency.

Concept: EM waves

SECTION B

17. Focal lengths (thin lens, Lens maker's formula)

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right).$$

For a plano–convex lens: one curved surface (radius R) and one plane ($R = \infty$):

$$\frac{1}{f_1} = (1.5-1)\frac{1}{R} = \frac{1}{2}\cdot\frac{1}{15} \Rightarrow f_1 = 30 \text{ cm}$$

For the plano–concave lens: the curved surface has the opposite sign, giving $f_2 = -30$ cm (So $f_1 = +30$ cm, $f_2 = -30$ cm.)

Image formed by lens 1(L_1)

Object at infinity $\Rightarrow$ image by lens 1 at its focal plane:

$$v_1 = +30 \text{ cm (to the right of lens 1)}$$

Lens 2(L_2) is 20 cm to the right of lens 1, so the image formed by lens 1 is located 30 − 20 = 10 cm to the right of lens 2. Thus, for lens 2 the incident rays represent a *virtual object* lying 10 cm on the right side of lens 2. Using the Cartesian sign convention this object distance is $u_2 = -10$ cm.

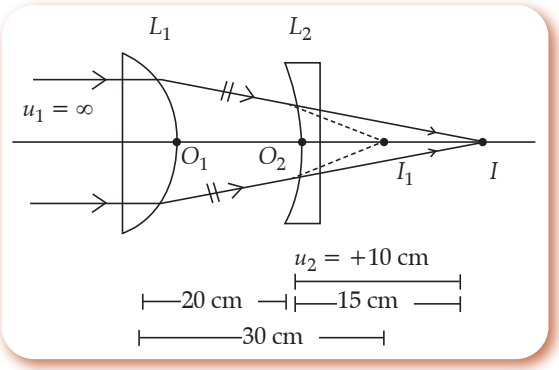
Image by lens 2 (use lens formula)

$$\frac{1}{f_2} = \frac{1}{v_2} + \frac{1}{u_2} \Rightarrow \frac{1}{-30} = \frac{1}{v_2} + \frac{1}{-10}.$$

So $$\frac{1}{v_2} = \frac{1}{-30} + \frac{1}{10} = \frac{-1+3}{30} = \frac{2}{30} = \frac{1}{15},$$

Hence

$$v_2 = +15 \text{ cm}.$$



Concept: Spherical lens

18. **(a) (i) Impact parameter (*b*):** The perpendicular distance between the initial straight-line path of the incoming α-particle (if it were undeflected) and the centre of the target nucleus.

Distance of closest approach: the minimum separation between the α-particle and the nucleus during the scattering process (the point where the radial separation is smallest).

(ii) Values for the impact parameter *b*:

- For scattering angle θ = 0° (no deflection) — $b \to \infty$ (effectively very large): the particle misses the nucleus by a large distance so it is undeflected.
- For scattering angle θ = 180° (complete back scattering) — $b = 0$: this corresponds to a head-on collision (trajectory aimed directly at the centre).

Concept: Impact parameter

OR

(b) Use Einstein's photoelectric equation:

$$K_{max} = h(\nu - \nu_0).$$

Given for frequency ν_1:

$$K = h(\nu_1 - \nu_0)$$

For ν_2: $$2K = h(\nu_2 - \nu_0)$$

Subtract/eliminate K: from the two equations,

$$\frac{K}{h} = v_1 - v_0, \quad \frac{2K}{h} = v_2 - v_0 \qquad [1]$$

$$\frac{K}{h} = v_2 - v_1 \Rightarrow \frac{K}{h} = v_2 - v_1$$

Now find v_0 (use $v_0 = v_1 - \frac{K}{h}$):

$$v_0 = v_1 - (v_2 - v_1) = 2v_1 - v_2. \qquad [1]$$

Concept: Einstein's photoelectric equation

19. The distance y from centre:

$$y = \left(\frac{D}{d}\right)\Delta x$$

where Δx is the path difference between two light waves reaching point P.

For the missing wavelengths (or minima points) at P,

$$\Delta x = \left(n + \frac{1}{2}\right)\lambda, \text{ where } n = 0, 1, 2, 3 \ldots$$

So $$y = \frac{D}{d}\frac{(2n+1)}{2}\lambda$$

$$\frac{d}{2} = \frac{D}{d}\frac{(2n+1)}{2}\lambda \quad \left(y = \frac{d}{2} \text{ from the figure}\right)$$

$$\lambda = \frac{d^2}{D(2n+1)}$$

for $$n = 0, 1, 2, 3, \ldots$$

So, $$\lambda = \frac{d^2}{D}, \frac{d^2}{3D}, \frac{d^2}{5D}, \ldots$$

will form minima at point P.

Concept: Young's double slit experiment

20. Due to the low charge and mass of the hydrogen nucleus compared to the alpha particles, we expect:

- Most alpha particles will pass through the solid hydrogen without any significant deflection.

- There will be very few instances of scattering, and those that do occur will be minimal.
- The results will not provide a clear picture of the nuclear structure, as the scattering angles will be very small and difficult to measure accurately.

Concept: Rutherford model of atom

21. **Given:**

- Doping ratio = 1 dopant atom per 5×10^{75} atoms
- Number density of Si atoms = 5×10^{28} atoms/m^3

Since there is 1 dopant atom per 5×10^{75} Si atoms:

Dopant number density

$$= \frac{\text{Number density of Si atoms}}{5\times10^{7}}$$

$$= \frac{5\times10^{28}}{5\times10^{7}} = 1\times10^{21} \text{ dopant atoms/m}^3$$

Convert to dopant atoms per cm^3

Since 1 m^3 = 10^6 cm^3:

$$\text{Dopant density} = \frac{1\times10^{21}}{10^{6}}$$

$$= 1 \times 10^{15} \text{ dopant atoms/cm}^3$$

Number of holes created per cm^3

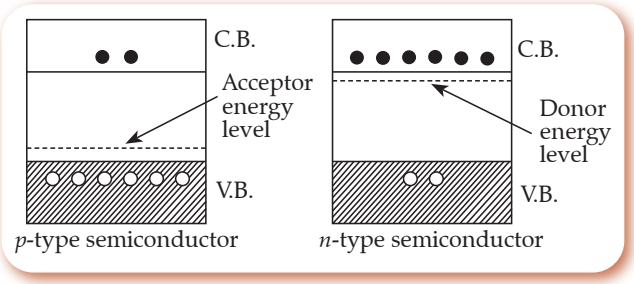
In a *p*-type semiconductor, each dopant atom (acceptor) creates one hole:

Number of holes = 1×10^{15} holes/cm^3

Example of dopant

- Example: Boron (B) is a common *p*-type dopant for silicon.

OR



[1] + [1]

Concept: Doping in semiconductor

SECTION C

22. **(a)** The kinetic energy of emitted photoelectrons varies because electrons within a metal have different initial energies before being ejected. According to Einstein's photoelectric equation:

$$K_{max} = h\nu - \phi$$

where:

- $h\nu$ is the energy of the incident photon,
- ϕ is the work function (minimum energy required to eject an electron),
- K_{max} is the maximum kinetic energy of the emitted photoelectron. [1]

Electrons located deeper inside the metal require more energy to escape than those near the surface, leading to a distribution of kinetic energies among the emitted photoelectrons.

(b) Saturation current refers to the maximum photoelectric current when all emitted photoelectrons are collected by the anode. Since, the intensity of light is directly proportional to the number of incident photons, a higher intensity means more photons striking the metal surface per second.

As each photon ejects at most one electron, an increase in intensity leads to a higher number of emitted electrons, resulting in an increased saturation current. However, the kinetic energy of individual electrons remains unaffected by intensity. [1]

(c) The photoelectric effect occurs only if the energy of the incident photons is at least equal to the metal's work function (ϕ). The energy of a photon is given by:

$$E = \frac{hc}{\lambda}$$

As the wavelength (λ) increases, the photon energy decreases. When the wavelength reaches a threshold value (λ_{th}), corresponding to the work function:

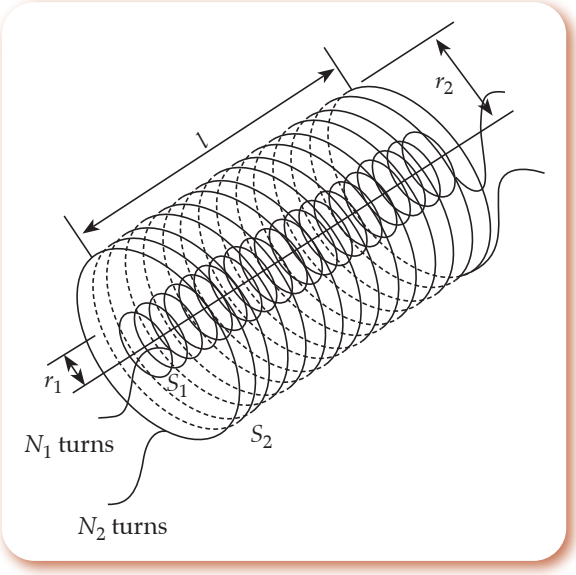
$$\lambda_{th} = \frac{hc}{\phi}$$

The photon energy becomes insufficient to eject electrons and photoemission ceases. Beyond this threshold wavelength, no electrons are emitted, regardless of the intensity of light. [1]

Concept: Photoelectric effect

23. Deriving the expression for mutual Inductance

Showing $M_{21} = M_{12}$



[½]

Let N_1 and N_2 be the total number of turns of coils S_1 and S_2 respectively.

When current I_2 is set up in S_2, flux linkage with solenoid S_1 is –

$$N_1\phi_1 = M_{12}I_2 \quad ...(i)$$

$$N_1\phi_1 = (n_1 l)(\pi r_1^2)(\mu_0 n_2 I_2)$$

$$N_1\phi_1 = \mu_0 n_1 n_2 \pi r_1^2 l I_2 \quad ...(ii)$$

From (i) and (ii)

$$M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l$$ [1½]

Considering reverse case, when I_1 current is set up in S_1, flux linkage with S_2 is –

$$N_2\phi_2 = M_{21}I_1 \quad ...(iii)$$

$$N_2\phi_2 = (n_2 l)(\pi r_1^2)(\mu_0 n_1 I_1) \quad ...(iv)$$

From (iii) and (iv)

$$M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l$$

$$\therefore \quad M_{12} = M_{21}$$ [1]

Concept: Mutual Inductance

24. **(a)** Magnetic field $\vec{B}$

According to Ampere's law

$$B = \frac{\mu_0 I}{2\pi d}$$

Direction is determined by right hand rule, magnetic field direction is along negative direction.

$$\vec{B} = \frac{-\mu_0 I}{2\pi d}\hat{j}$$ [1]

(b) Magnetic force $\vec{F}_B$

Charge is moving along $-x$ direction

so $$\vec{v}\times\vec{B} = -(v\hat{i})\times\left(-\frac{\mu_0 I}{2\pi d}\hat{j}\right)$$

$$\vec{F}_B = qv\left(-\frac{\mu_0 I}{2\pi d}\hat{k}\right)$$

So direction of magnetic force is negative. [1]

(c) Electric field $\vec{E}$

Charge is moving undeviated

$$\vec{F}_E + \vec{F}_B = 0$$

$$\vec{F}_E = q\vec{E}$$

$$\Rightarrow \quad q\vec{E} = -\left(-\frac{qv\mu_0 I}{2\pi d}\hat{k}\right)$$

$$\vec{E} = \frac{v\mu_0 I}{2\pi d}\hat{k}$$ [1]

Concept: Moving charge electric and magnetic field

25. **Topper's Answer 2023**

At resonance, I_m is maximum.

$I_m = \frac{V_m}{Z} \Rightarrow$ Z should be minimum

$Z = \sqrt{R^2 + (X_L - X_C)^2}$, Z will be minimum when $X_L = X_C$, i.e, $Z_{min} = R$

$X_L = X_C$

$L\omega_0 = \frac{1}{C\omega_0} \Rightarrow \omega_0^2 = \frac{1}{LC}$

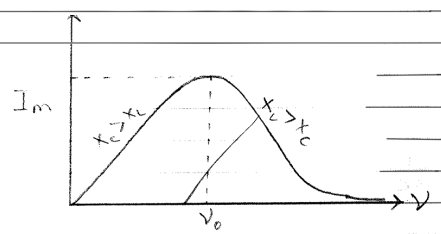
$\omega_0 = \frac{1}{\sqrt{LC}}$, where ω_0 is the resonant angular frequency

$\nu = \frac{\omega}{2\pi} \Rightarrow \nu_0 = \frac{1}{2\pi}\frac{1}{\sqrt{LC}} \Rightarrow \nu_0 = \frac{1}{2\pi\sqrt{LC}}$, where ν_0 is the resonant frequency

→ ν_0 (resonant frequency) depends on :

- Inductance of the inductor
- Capacitance of the capacitor

→



Concept: LCR circuit

26. **(a)** Three key experimental results that could not be explained by the wave theory:

1. Instantaneous Emission of Photoelectrons: Electrons are emitted immediately when light strikes the metal surface, without any delay, no matter how weak the light source is. According to wave theory, energy should be gradually absorbed over time, causing a delay in electron emission, which does not happen.

2. Existence of a Threshold Frequency: Photoelectrons are only emitted if the light frequency f is above a certain threshold frequency f_0, regardless of the intensity of light. Wave theory suggests that a higher intensity (brighter light) should provide enough energy for electrons to escape, even at low frequencies. However, no electrons are emitted below f_0, contradicting wave theory.

3. Kinetic Energy of Electrons Depends on Frequency, Not Intensity: The maximum kinetic energy of emitted electrons increases with the frequency of light but is independent of intensity. Wave theory predicts that increasing light intensity should increase energy transfer to electrons, increasing their kinetic energy. Instead, only higher frequencies result in higher kinetic energy.

(b) Slope of cut off voltage (V_s) V_s frequency is given

$$\text{Slope} = 4.12 \times 10^{-15}\ \text{Vs}$$

Einstein photoelectric equation

$$eV_0 = hf - \phi$$

or $$V_0 = \frac{hf}{e} - \frac{\phi}{e}$$

also $$\text{slope} = \frac{h}{e}$$

$$\therefore \quad \frac{h}{e} = 4.12 \times 10^{-15}\ \text{Vs}$$

$$h = 4.12 \times 10^{-15} \times 1.6 \times 10^{-19}$$

$$= 6.592 \times 10^{-34}\ \text{Js}$$

$$h = 6.63 \times 10^{-34}\ \text{Js}$$

Concept: Combination of Lenses

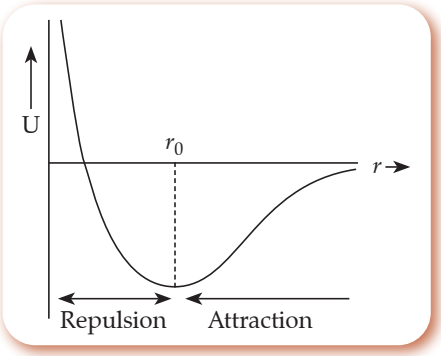
27. **(i)** $$\rho = \frac{\text{mass}}{\text{volume}}$$

$$= \frac{\text{mass number} \times \text{mass of nucleon}}{\text{volume of nucleus}}$$

$$\rho = \frac{A \times m}{\frac{4}{3}\pi(R_0 A^{\frac{1}{3}})^3} = \frac{3m}{4\pi R_0^3}$$

Hence, density is independent of mass number.

(ii)

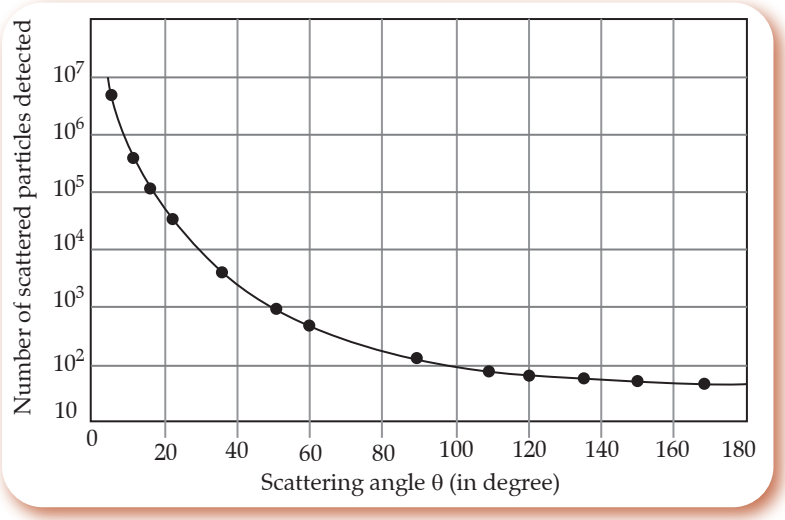


Inferences

- The force is attractive for distances larger than r_0.
- The force is repulsive for distance less than r_0.

OR

(b) (i)



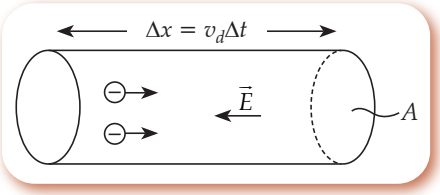
(ii) –The entire positive charge and most of the mass of the atom are concentrated in a small space.

–Most α-particles pass through the foil. It means that they do not suffer any collisions.

To deflect the α-particle backwards, a large repulsive force is required, which is possible if the mass of the atom & positive charge of the atom are concentrated tightly at its centre. This lead to the discovery of the nucleus in the atom.

Concept: Nucleus

28. **(i)**



[1]

Current in the conductor

$$I = neAv_d$$

$$\frac{I}{A} = ne\frac{eE}{m}\tau$$

$$J = \frac{ne^2\tau}{m}E$$

$$J = \frac{1}{\rho}E$$

$$\rho = \frac{m}{ne^2\tau}$$ **[1]**

(ii) From given graph

$$\frac{Q_1}{Q_2} = \frac{A_1(\text{Area of rectangle})}{A_2(\text{Area of triangle})}$$

$$\frac{Q_1}{Q_2} = \frac{2}{\frac{3}{2}}$$

$$Q_1 > Q_2$$ **[1]**

Concept: Electric current

SECTION D

29. **(i) (d)** 6

Why? Let charge on $A = Q$ (given).

Top-left node: $q_B + Q = 0 \Rightarrow q_B = -Q$

Top-right node: $q_M - Q = 0 \Rightarrow q_M = Q$.

Voltages:

$$V_{TL} = -\frac{Q}{C}$$

$$V_{TR} = V + \frac{Q}{C}$$

Across A:

$$V_{TL} - V_{TR} = \frac{Q}{C} \Rightarrow -\frac{Q}{C} - \left(V + \frac{Q}{C}\right)$$

$$= \frac{Q}{C} \Rightarrow Q = \frac{CV}{3}$$

N is across battery: $Q' = 2CV$.

Ratio:

$$\frac{Q'}{Q} = \frac{2CV}{\frac{CV}{3}} = 6.$$

OR

(ii) (c) $\frac{1}{2}\varepsilon_0 E^2 V$

Why? Energy density in an electric field

$$u = \frac{1}{2}\varepsilon_0 E^2$$

Total energy in volume V

$$U = u \cdot V$$

$$U = \frac{1}{2}\varepsilon_0 E^2 \cdot V$$

(iii) (b) $\frac{\sigma}{\sigma - \sigma_p}$

Why? Field in vacuum region (between plate and slab)

$$E_{vac} = \frac{\sigma}{\varepsilon_0}$$

Field inside the dielectric

$$E_{in} = \frac{(\text{free surface charge on plate} - \text{bound charge on slab face})}{\varepsilon_0}$$

$$E_{in} = \frac{\sigma - \sigma_p}{\varepsilon_0}$$

Relation between E_{vac} and E_{in}

$$E_{in} = \frac{E_{vac}}{K}$$

Substitute and solve

$$\frac{\sigma - \sigma_p}{\varepsilon_0} = \frac{1}{K} \cdot \frac{\sigma}{\varepsilon_0}$$

$$K = \frac{\sigma}{\sigma - \sigma_p}$$

(iv) (a) $\frac{C}{4}$

Why? Initial capacitor

$$C = \frac{\varepsilon_0 A}{d}$$

New overlap area $= \frac{A}{2}$, new separation $= 2d$.

New capacitance

$$C_{new} = \frac{\varepsilon_0\left(\frac{A}{2}\right)}{2d} = \frac{\varepsilon_0 A}{4d} = \frac{C}{4}$$

Concept: Capacitance

30. **(i)** We have photoelectric equation

$$\text{K.E.} = hv - h\nu_0$$

or $$eV_0 = h\nu - h\nu_0$$

we can write it as

$$V_0 = \frac{h}{e}\left(\frac{\nu}{e} - \nu_0\right)$$

or $$V_0 = \frac{h}{e}\left(\frac{\nu}{e} - \nu_0\right)$$

on comparing it with standard straight line equation

$$y = mx + C$$

we get

intercept on v-axis : v_0 [1]

intercept on V_0-axis : $\frac{hV_0}{e}$ [1]

(ii) Frequency of photon is given as:

$$f = \frac{C}{2\lambda} = \frac{1}{2}f$$

if wavelength is doubled, then new frequency,

$$f' = \frac{1}{\lambda}$$ [½]

Now, wave number (k) is given as

$$k = \frac{1}{2\lambda} = \frac{1}{2}k$$ [½]

New wave number (k') will be

(iii) Wavelength is given by

$$\lambda = \frac{h}{p} = \frac{6.626\times10^{-34}}{5.0\times10^{-29}} = 13.25\times10^{-6}\,\text{m}$$ [1]

Concept: Photoelectric effect

SECTION E

31. **(a) (i) Near point *P***

Magnetic field is acting into the plane of the paper as force is acting upwards.

Near point *Q*

Magnetic field is into the plane of paper as force is acting upwards.

Near point *R*

Magnetic field is acting out of the plane of the paper as $\vec{F}$ is acting downwards. [½] + [½] + [½]

Relative magnitude of the magnetic field:

As $B \propto \frac{1}{r}$

Therefore,

Near point P, magnitude of B is small.

Near point Q, B is relatively smaller than point P.

Near point R, B is relatively larger than point P.

($B_Q < B_P < B_R$) [½] + [½] + [½]

(ii) Let r be the radius of the circular coil and I is the current in the coil, then

$$B = \frac{\mu_0 I}{2r} \text{ or } I = \frac{2Br}{\mu_0}$$

$$A = \pi r^2 ; r = \sqrt{\frac{A}{\pi}}$$

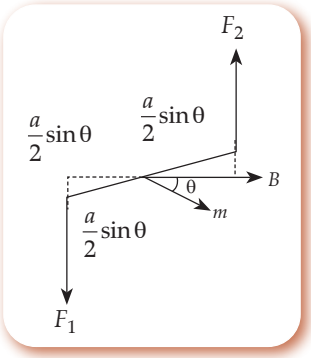
$$M = IA$$

$$= \frac{2Br}{\mu_0}A$$

$$= \frac{2Br}{\mu_0}\sqrt{\frac{A}{\pi}}$$ [2]

OR

(b) (i)



$\vec{F_1}$ and $\vec{F_2}$ are the forces acting on two arms of the rectangular coil having sides a and b.

$$|\vec{F_1}| = |\vec{F_2}| = IbB \qquad (b = \text{length of the arm})$$

Forces constitute a couple. The magnitude of Torque on the loop is –

$$\tau = F_1\frac{a}{2}\sin\theta + F_2\frac{a}{2}\sin\theta$$

$$= I\,a\,b\,B\sin\theta$$

$$= IAB\sin\theta$$

$$\vec{\tau} = I\vec{A}\times\vec{B}$$

(ii) (1) $$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

$$r \propto \sqrt{K}$$

$$\frac{r'}{r} = \frac{\sqrt{K/2}}{\sqrt{K}} = \frac{1}{\sqrt{2}}$$

$$r' = \frac{r}{\sqrt{2}}$$

(2) $$T = \frac{2\pi m}{qB}$$

Time period does not depend on kinetic energy.

$\therefore$ Time period will not change.

Concept: Magnetism

32. **(a) (i)** Given, a thin pencil of length $\frac{f}{4}$ is placed along the principal axis of a mirror of focal length f.

The image of the pencil is real and enlarged, meaning the mirror must be a concave mirror with the object placed between f and $2f$ to ensure that its image forms beyond $2f$ and just touches suggests that the image coincides with one end of the pencil.

Let the object be placed at distance u from the mirror, forming an image at v. The magnification is given by: $m = -\frac{v}{u}$.

We assume the object is placed at $u = 1.5f \left(\text{or } \frac{3f}{2}\right)$ and it is an enlarged and real image.

By the mirror formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Substituting $u = \frac{3f}{2}$

$$\frac{1}{f} = \frac{1}{\frac{3f}{2}} + \frac{1}{v}$$

$$\frac{1}{f} - \frac{2}{3f} = \frac{1}{v}$$

$$\frac{(3-2)}{3f} = \frac{1}{v}$$

$$\frac{1}{3f} = \frac{1}{v}$$

$$v = 3f$$

Magnification: $m = \frac{-v}{u} = \frac{-3f}{\frac{3f}{2}} = -2$

Since, magnification is negative, the image is inverted and twice the size of the object.

(ii) Given, angle of incidence $i = 45°$, angle of deviation, $\delta = 15°$, angle of prism, $A = 30°$.

Using the prism formula: $\delta = i + e - A$

Substituting the given values:

$$15° = 45° + e - 30°$$

$$e = 0°$$

Since, the angle of emergence (e) is zero, the emergent ray is normal to the face AC.

For refractive index by using Snell's law at face AB:

$$n_1 \sin i = n_2 \sin r$$

Since the incident ray is in air, $n_1 = 1$ and let $n_2 = n$ be the refractive index of the prism material. The angle of refraction at face AB is denoted by r.

$$\sin 45° = n \sin r$$

$$\frac{\sqrt{2}}{2} = n \sin r \quad \text{...(i)}$$

Using the prism angle formula: $r + r' = A$

$$r + r' = 30°$$

Since, the emergent ray is normal to AC, $r' = 0°$

So, $r = 30°$

Now, substituting $r = 30°$ in Eq. (i)

$$\frac{\sqrt{2}}{2} = n \sin 30°$$

$$\frac{\sqrt{2}}{2} = n \times \frac{1}{2}$$

$$n = \sqrt{2} \approx 1.414$$

Concept: Ray optics

OR

(b) (i) Given

Wavelengths: $\lambda_1 = 600$ nm $= 6 \times 10^{-7}$ m and $\lambda_2 = 480$ nm $= 4.8 \times 10^{-7}$ m

Distance between slits: $d = 1.0$ mm $= 1.0 \times 10^{-3}$ m

Distance from slits to screen: $L = 1.0$ m

(1) Distance of the third bright fringe for $\lambda_1 = 600$ nm

The fringe width

$$y = \frac{m\lambda L}{d}$$

For the third bright fringe ($m = 3$):

$$y_3 = \frac{3 \times (6 \times 10^{-7}) \times 1}{1.0 \times 10^{-3}}$$

$$y_3 = \frac{3 \times 6 \times 10^{-7}}{10^{-3}}$$

$$y_3 = \frac{18 \times 10^{-7}}{10^{-3}} = 1.8 \text{ mm}$$

(2) Least distance from the central maximum where bright fringes coincide:

Bright fringes from both wavelengths will coincide at a position where the order numbers m_1 and m_2 satisfy:

$$m_1\lambda_1 = m_2\lambda_2$$

The LCM of the wavelengths gives the least position where both fringes coincide, i.e., LCM

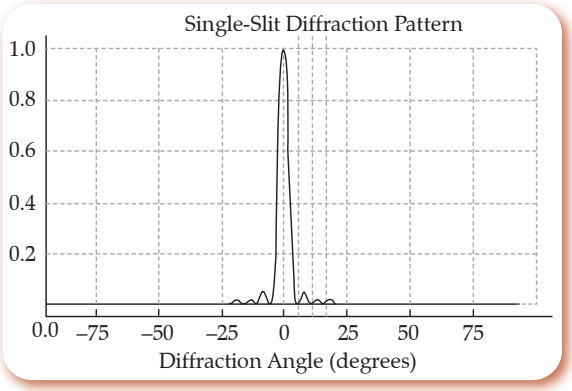
$(600, 480) = 2400$ nm $= 2.4 \times 10^{-6}$ m.

Now, the fringe distance for this position:

$$y = \frac{2.4 \times 10^{-6} \times 1}{10^{-3}}$$

$$y = \frac{2.4 \times 10^{-6} \times 1}{10^{-3}} = 2.4 \text{ mm}$$

(ii) (1) Graph of Intensity vs. Angle of Diffraction



The dark fringes (minima) in a single-slit diffraction occur at angles where destructive interference happens. These angles are given by:

$$a \sin \theta = m\lambda$$

where: a = slit width, λ = wavelength of light, θ = diffraction angle, m = 1, 2, 3, ... (integer, not including 0, since $m = 0$ corresponds to the central maximum)

(2) Difference between Diffraction of Light and Sound Waves:

Feature	Light Waves	Sound Waves
Wavelength	Very small ($\sim 10^{-7}$m)	Larger ($\sim 10^{-3}$ to metres)
Diffraction observability	Requires very narrow slits/ gaps	Easily observed around obstacles
Practical example	Light bending around a sharp edge	Sound heard around corners
Extent of diffraction	Less noticeable due to small wavelength	More noticeable due to larger wavelength

Thus, sound waves diffract more easily around obstacles compared to light waves due to their longer wavelengths.

Concept: Diffraction

33. **(a) (i)** Given

$$\vec{E} = \frac{3\times10^5}{r^2}\hat{r}$$

$$V = \frac{3\times10^5}{r}$$ **[1]**

Electrostatic energy of the system in the absence of the field

$$U_i = \frac{Kq_1q_2}{r_{12}}$$ **[1]**

Electrostatic energy in the presence of the field

$$U_f = \frac{Kq_1q_2}{r_{12}} + q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$$

$$\Delta U = U_f - U_i = q_1V(\vec{r}_1) + q_2V(\vec{r}_2)$$

$$\Delta U = \frac{5\times10^{-6}\times3\times10^5}{3\times10^{-2}} - \frac{1\times10^{-6}\times3\times10^5}{3\times10^{-2}}$$

$$= 40 \text{ J}$$ **[2]**

(ii) (1) $C = \frac{Q}{V} = \frac{80\times10^{-3}}{16} = 5.0\times10^{-3}\text{F} = 5 \text{ mF}$.

(2) With dielectric $K = 3$, $C^2 = KC$. For same charges,

$$V' = \frac{Q}{C'} = \frac{V}{K} = \frac{16}{3}\text{V} \approx 5.33 \text{ V}$$

(3) No, the capacitance of the system depends on its geometry. **[1]**

OR

(b) (i) Total charge for A = Total charge for B = Total charge for $C = +4q$

As, $$E = \frac{kQ}{r^2}$$

Since $Q = 4q$ and $r = 3R$

$$E = \frac{k(4q)}{9R^2} = \frac{4kq}{9R^2}$$

$\therefore$ $E_A = E_B = E_C$ **[2]**

(ii) $$V_C = \left[\frac{k\times6\times10^{-6}}{5\times10^{-2}} - \frac{k\times6\times10^{-6}}{5\times10^{-2}}\right] = 0$$

$$V_A = \left[\frac{k\times6\times10^{-6}}{15\times10^{-2}} - \frac{k\times6\times10^{-6}}{5\times10^{-2}}\right]$$

$$= \frac{k\times6\times10^{-6}}{10^{-2}}\left[\frac{1-3}{15}\right]$$

$$= -\frac{9\times10^9\times6\times10^{-6}\times2}{15\times10^{-2}}$$

$$= -7.2\times10^5 \text{ V}$$

$$W = q[V_A - V_C]$$

$$= 5\times10^{-6}[-7.2\times10^5 - 0]$$

$$W = -3.6 \text{ J}$$ **[3]**

Concept: Electrostatic